# Band-1 Computational Experiment: Oppermann's Conjecture — Finite Split-Interval Verification

Human + ZoraASI full-dash (executor)

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## Band-1 Computational Experiment: Oppermann's Conjecture (Finite Verification)

**Honesty stamp.** This note **verifies** that for every integer $n$ with $2 \le n \le N$ the open intervals $(n(n-1), n^2)$ and $(n^2, n(n+1))$ each contain at least one prime, for an explicit finite $N$ computed on this box. It does **not** prove **Oppermann's conjecture** for all $n$. Finite-range verification $\neq$ conjecture. **Status: OPEN.** This is **not** a Clay Millennium claim and **not** a ToE / GATE-04 unlock.

### Abstract

We check both Oppermann half-intervals abutting each square $n^2$ for $n = 2, \dots, N$ using `gmpy2.is_prime` (Baillie–PSW) on odd candidates, with an odds-bit Eratosthenes cross-check on a smaller range, and obtain

$$\mathbf{N = 100{,}000 = 10^5}, \qquad \mathbf{failures = 0}$$

(primary BPSW wall **0.90 s**). Sieve cross-check at $N = 10^4$: **0** failures (wall **17.58 s**). **Conjecture status: OPEN.** Lit record $\gg$ box ($n \le 7.05 \cdot 10^{13}$).

### Problem statement

**Oppermann's conjecture** (1882). For every integer $n > 1$ there exists a prime $p$ with

$$n(n-1) < p < n^2$$

and a prime $q$ with

$$n^2 < q < n(n+1).$$

**Implications (logical).** Oppermann strictly implies **Legendre's conjecture** (a prime in $(n^2, (n+1)^2)$) and yields Bertrand-type strengthenings near consecutive squares (primes guaranteed in both length-$\sim n$ windows abutting $n^2$). These implications are **not** a proof of Oppermann.

**This experiment.** Finite verification for $n = 2..N$ on this box; literature ledger only for larger published ranges.

### Literature ledger (not re-proved on box)

| Year | Result | Authors | Status |
|---|---|---|---|
| 1882 | Oppermann's conjecture stated | Oppermann | historical |

| Year | Result | Authors | Status |
|---|---|---|---|
| $ $2014 | Oppermann/Legendre to $n \le 2 \cdot 10^9$ | via Oliveira e Silva–Herzog–Pardi tables | prior computational bound |
| **2024/2025** | **Oppermann for all** $n \le 7.05 \cdot 10^{13}$ | **Sorenson & Webster**, *Res. Number Theory* **11** | **published record** |

Primary citation: J. Sorenson, J. Webster, *An algorithm and computation to verify Legendre's conjecture up to* $7 \cdot 10^{13}$, Research in Number Theory **11** (2025); DOI 10.1007/s40993-024-00589-4 (verifies **Oppermann**, hence Legendre).

**Unconditional Oppermann for all $n$ remains OPEN.**

## Method (box)

1. **Unit tests.** $n = 2, 3, 10$; BPSW verify $n = 2..80$; sieve verify $n = 2..500$; sieve BPSW agreement on $n = 2..200$.
2. **Sieve cross-check.** Odds-only bit-packed Eratosthenes to $N(N+1)$ for $N = 10^4$; scan both open intervals.
3. **Primary BPSW.** For each $n = 2..10^5$, scan odd candidates in $(n(n-1), n^2)$ and $(n^2, n(n+1))$ with `gmpy2.is_prime` until a prime is found (or fail).
4. **Failure log.** Expect **none**. Full sieve to $N = 10^5$ ($L \sim 10^{10}$) was aborted as too slow for the dash budget; primary uses BPSW at $N = 10^5$.

## Results (numbers!)

### Unit tests

`PASS`: $(2, 4) \ni 3$, $(4, 6) \ni 5$; $(6, 9) \ni 7$, $(9, 12) \ni 11$; $n = 10$ both sides; BPSW $n = 2..80$ zero failures; sieve $n = 2..500$ zero failures; sieve/BPSW agree $n = 2..200$.

### Box verification

| Quantity | Value |
|---|---|
| **Primary $N$ (BPSW)** | $100{,}000 = 10^5$ |
| Integers $n$ checked | 99,999 ($n = 2..N$) |
| **Failures (left / right / total)** | **0 / 0 / 0** |
| All verified | **True** |
| Max left offset above $n(n-1)$ | **221** at $n = 81{,}070$ |
| Max right offset above $n^2$ | **225** at $n = 87{,}598$ |
| Mean left / right offset | 19.2950 / 18.7175 |
| Primary wall | **0.8972 s** |
| Sieve cross-check $N$ | 10,000 |
| Sieve failures | **0** |
| Sieve wall | **17.5767 s** |
| Total wall (tests + both) | **18.5139 s** |

### Sample primes found

| $n$ | $n^2$ | left prime in $(n(n-1), n^2)$ | right prime in $(n^2, n(n+1))$ |
|---|---|---|---|
| 2 | 4 | 3 | 5 |
| 3 | 9 | 7 | 11 |

| $n$ | $n^2$ | left prime in $(n(n-1), n^2)$ | right prime in $(n^2, n(n+1))$ |
|---|---|---|---|
| 10 | 100 | 97 | 101 |
| 100 | $10^4$ | 9901 | 10007 |
| 1{,}000 | $10^6$ | 999007 | 1000003 |
| 10{,}000 | $10^8$ | 99990001 | 100000007 |
| 100{,}000 | $10^{10}$ | 9999900017 | 10000000019 |

## What this does not claim

- Oppermann's conjecture for all $n$ (still **OPEN**).
- Legendre / Andrica / Brocard proved.
- Clay Millennium / ToE / GATE-04 unlock.
- Matching the Sorenson–Webster $7.05 \cdot 10^{13}$ record on this box.

## Reproducibility

```
python3 experiments/band1/oppermann/oppermann_verify.py
# writes results.json, literature_ledger.json, SUMMARY.txt
```

Requires: Python 3, `gmpy2`.

## Conclusion

For all $n = 2, \ldots, 10^5$, both Oppermann intervals contain a prime (**0** failures; BPSW wall $\sim 0.9$ s). Sieve cross-check $n = 2..10^4$ also **0** failures. Literature verifies much farther ($n \leq 7.05 \cdot 10^{13}$). **Oppermann's conjecture remains OPEN.**